04-12 第二篇

The Science of Origami: From Paper Folds to Space

When you think of origami, you might picture a paper crane or a folded frog. But this ancient Japanese art is no longer just a hobby—engineers, doctors, and even NASA scientists are using origami principles to solve real-world problems. [1] The key is that folding allows flat materials to expand, shrink, or change shape in clever ways without adding extra parts. A folded structure can be packed small for travel and then opened up when it is needed. This makes origami ideal for situations where space and weight matter, such as in spacecraft, medical devices, or disaster relief.

Space is one of the most exciting applications. Satellites and telescopes need large solar panels or antennas to work well, but rockets have limited room inside them. [2] Engineers have designed solar panels that fold up like a fan or an accordion so they fit inside the rocket during launch. Once in orbit, they unfold automatically and spread out to catch sunlight. Some designs are inspired by origami patterns that have been studied for centuries. Without folding, we could never send such large structures into space efficiently.

Medicine is another field where origami is making a difference. [3] Surgeons sometimes need to insert tools or implants through very small cuts to reduce scarring and healing time. A stent—a tube that holds open a blocked blood vessel—can be folded flat, pushed through a narrow catheter, and then expanded inside the body. The same idea is used for heart implants and even tiny robots that could one day move inside the human body to deliver medicine. Folding lets big devices work through small openings.

Origami also helps in emergencies. [4] Temporary shelters for people who have lost their homes in disasters can be made from flat panels that fold into 3D structures. These shelters are easy to transport and can be set up quickly without heavy machinery. In some places, foldable solar panels and water-purification devices use the same principle—pack small, deploy big. When every kilogram and cubic meter counts, origami designs save both space and time.

[5] From space missions to life-saving medical tools and disaster relief, the simple act of folding paper has led to inventions that reach far beyond the craft table. The next time you fold a paper plane, remember that you are using the same principles that engineers use to explore the universe.

选项

A. In fact, origami-based designs are now used in satellites, medical implants, and emergency shelters. B. The solution is to use origami: fold everything down for the

journey, then unfold it in space. C. Here, the challenge is to fit large instruments into a small launch vehicle. D. Folding allows devices that would be too big to insert through a small cut to be compressed and then expanded where they are needed. E. In these cases, foldable structures can be shipped flat and assembled quickly on site. F. All in all, origami proves that a centuries-old art can inspire cutting-edge technology in surprising ways.

答案键（Answer Key）

[1] A [2] B [3] D [4] E [5] F

深度解析

- **[1] A**：
  - **线索**：原理总述与应用领域引出。空前解释"folding allows flat materials to expand, shrink...packed small for travel...opened up when needed"并指出"ideal for situations where space and weight matter"。A选项"In fact, origami-based designs are now used in satellites, medical implants, and emergency shelters"（事实上，基于折纸的设计现已用于卫星、医疗植入物和应急避难所）中"In fact"承接原理阐述，"satellites, medical implants, emergency shelters"与空后"spacecraft, medical devices, disaster relief"一一对应，自然引出下文三大应用领域，衔接顺畅。
- **[2] B**：
  - **线索**：太空难题与折纸解决方案。空前提出"Satellites and telescopes need large solar panels or antennas...but rockets have limited room inside them"，空后具体描述"Engineers have designed solar panels that fold up like a fan...fit inside the rocket...Once in orbit, they unfold automatically"。B选项"The solution is to use origami: fold everything down for the journey, then unfold it in space"（解决方案是使用折纸：在旅程中将一切折叠收起，到太空后再展开）中"The solution"直接回应空前难题，"fold...for the journey, then unfold in space"与空后"fold up...fit inside...unfold automatically...in orbit"完美对应，逻辑紧密。
- **[3] D**：
  - **线索**：医疗应用与折叠机制。空前指出"Surgeons sometimes need to insert tools or implants through very small cuts"，空后举例"stent...can be folded flat, pushed through a narrow catheter, and then expanded inside the body...heart implants...tiny robots"。D选项"Folding allows devices that would be too big to insert through a small cut to be compressed and then expanded where they are needed"（折叠使本因过大而无法通过小切口插入的设备得以压缩，并在需要处展开）中"too big to insert through a small cut"与空前"very small cuts"呼

应，"compressed and then expanded where they are needed"与空后"folded flat...pushed through...expanded inside the body"精准对应，衔接自然。

- **[4] E**：
  - **线索**：应急场景与可折叠结构优势。空前提出"Temporary shelters for people who have lost their homes...can be made from flat panels that fold into 3D structures"，空后说明"easy to transport...set up quickly without heavy machinery...foldable solar panels and water-purification devices...pack small, deploy big"。E选项"In these cases, foldable structures can be shipped flat and assembled quickly on site"（在这些情况下，可折叠结构可以平装运输并在现场快速组装）中"In these cases"指代应急场景，"shipped flat"与空后"flat panels...easy to transport...pack small"对应，"assembled quickly on site"与"set up quickly...deploy big"呼应，逻辑顺畅。
- **[5] F**：
  - **线索**：全文收束与升华。空[5]位于结尾段段首，空后列举"space missions, life-saving medical tools, disaster relief"并总结"simple act of folding paper has led to inventions...principles that engineers use to explore the universe"。F选项"All in all, origami proves that a centuries-old art can inspire cutting-edge technology in surprising ways"（总之，折纸证明一门古老的艺术能以令人惊讶的方式启发前沿技术）用"All in all"总结全文，"centuries-old art"与"simple act of folding paper"呼应，"cutting-edge technology"与"space missions...medical tools...explore the universe"形成升华，收束得当。
- **干扰项分析**：
  - **C. The challenge is to fit large instruments into a small launch vehicle.**：描述太空任务中"大仪器装入小火箭"的难题，易被误填[2]。但空[2]空前已明确说明"Satellites and telescopes need large...but rockets have limited room"，C merely 重复难题而未提供解决方案；空后需引出"fold up...unfold"的折纸解法，B的"The solution is to use origami"直接给出答案并衔接空后设计描述。C属"重复前文、缺乏推进"型干扰。